Q1 — Which gate (2-input OR, AND, XOR) produces the most "unpredictable" (highest entropy) output, assuming inputs are uniformly random over {00,01,10,11}?

Step 1 — compute output distributions

Inputs are equally likely (4 equally likely pairs).

- **OR**: outputs
  OR(00)=0, OR(01)=1, OR(10)=1, OR(11)=1 →
  $P(Y=0)=1/4,\ P(Y=1)=3/4.$
- **AND**: outputs
  AND(00)=0, AND(01)=0, AND(10)=0, AND(11)=1 →
  $P(Y=0)=3/4,\ P(Y=1)=1/4.$
- **XOR**: outputs
  XOR(00)=0, XOR(01)=1, XOR(10)=1, XOR(11)=0 →
  $P(Y=0)=2/4=1/2,\ P(Y=1)=1/2.$

Step 2 — compute Shannon entropy $H=-\sum p\log_2 p$

- For OR (and similarly for AND since they are symmetric):
  $H=-\big(0.25\log_2 0.25+0.75\log_2 0.75\big)$
  $\log_2 0.25=-2,\ \log_2 0.75\approx -0.415037.$
  So $H\approx 0.25\cdot 2+0.75\cdot 0.415037\approx 0.5+0.311278=0.811278$ bits.
- For XOR: $P=0.5, 0.5\to H=-2(0.5\log_2 0.5)=1$ bit.

Conclusion / intuition

- **XOR** has the highest entropy: **1 bit** (most unpredictable).
  Why: XOR is *balanced* under uniform inputs (outputs 0 and 1 equally likely), so the output is a fair coin.
- **OR** and **AND** both have the same, lower entropy = **0.8113 bits**, because they are biased (one output value is more likely than the other).

So answer: **XOR** produces the most informative / unpredictable output.

CAT-1 Preparations

ITA Tutorial Sessions for Unit-I & Unit-II

Date: 11 ur)

Q2 — Dice throwing experiment: you can set the outcome probabilities before the throw. How many ways to achieve maximum and minimum entropy, and what are those entropies?

We have 6 faces. Let distribution be $p=(p_1,\dots,p_6)$, $p_i\ge 0$, $\sum p_i=1$.

- **Maximum entropy** occurs when the distribution is **uniform**: $p_i=1/6$ for all i. There is exactly **one** probability vector of that form (though you can permute labels, it's the same distribution).
  Entropy: $H_{\max}=\log_2 6\approx 2.584962$ bits.
- **Minimum entropy** occurs when the distribution is **degenerate** (one face has probability 1, others 0). There are **6** such degenerate distributions (one for each face).
  Entropy: $H_{\min}=0$ bits.

Summary: Max: uniform (1 way), $H\approx 2.585$ bits. Min: any deterministic face (6 ways), $H=0$.

Unit-I

Q3 — Train arrives uniformly in [2:50, 3:10]. Probability it arrives between 3:00 and 3:01?

Total window length = 20 minutes (from 2:50 to 3:10). Target interval length = 1 minute (3:00–3:01). For a uniform continuous distribution:

$$P=\frac{1}{20}=0.05=5\%.$$

(If you insist on including endpoints exactly it doesn't change the result for continuous uniform.)

Special Tutorial - I

Unit-II

1. Consider three types of logic gates: a 2-input OR gate, AND gate, and XOR gate. Given that each input state (00, 01, 10, 11) is equally likely, which of these gates produces an output that is more "unpredictable" or "informative" in terms of entropy? Can you explain why this is the case?"

2. Let us consider the dice throwing experiment. Assume the probability of the events can be set before the throw. In how many ways you can achieve the maximum and minimum entropy, and what are they?

Unit-I

3. Suppose you are waiting for a train at a train station. According to the train schedule, the train is expected to arrive at 3:00 PM with a time window of ±10 minutes. That means the train could arrive anytime between 2:50 PM and 3:10 PM. What is the probability that the train will arrive between 3:00 PM and 3:01 PM?

4. You are at a concert where the lights are scheduled to dim at exactly 9:00 PM. However, the timing can vary slightly, and the lights are expected to dim at 9:00 ± 3 minutes. This means the lights can dim any time between 8:57 PM and 9:03 PM. What is the probability that the lights will dim exactly at 9:00 PM?

5. You have a bag with 3 colored balls: one red, one blue, and one green. Each ball has an equal chance of being picked.
   - The red ball is worth 1 point.
   - The blue ball is worth 2 points.
   - The green ball is worth 3 points.

   Determine the following:
   i. What is the arithmetic mean of the points assigned to the balls?
   ii. What is the expected value (mean) of the points if you pick one ball at random?

6. You have a simple game where you roll a fair six-sided die. Each face of the die shows a different number from 1 to 6. In this game, the number you roll corresponds to the points you win.
   Determine the following:

i. What is the arithmetic mean of the points on the die?
ii. What is the expected value (mean) of the points you will win if you roll the die?

7. In a class, 60% of the students have completed their assignment. Three students are selected at random. It is known that at least one of them has completed their assignment. What is the probability that all three students have completed their assignment?

8. In a video game, 70% of the players successfully complete a level on their first attempt. Five players are randomly selected, and it is known that at least one of them successfully completed the level on their first attempt. What is the probability that all five players completed the level on their first attempt?

9. You are at a party where it's known that 40% of the attendees are wearing hats. You randomly spot three people at the party. It is known that the first person you spot is wearing a hat. What is the probability that the other two people are also wearing hats?

10. You have a lamp that can be turned on by either of two independent switches, S1 and S2. Each switch can be either ON or OFF, and each switch is equally likely to be in either state. The switches are connected in parallel to the lamp. What is the probability that the lamp will be on?

Q4 — Lights dim uniformly in [8:57, 9:03] (±3 min). Probability they dim exactly at 9:00?

For a continuous uniform distribution the probability of any single exact instant (a single point) is **0** (measure zero).
So the answer is **0** (practically zero).
(If you meant "within ±1 second" or a small interval, replace the point by that interval and use length/total length.)

Q5 — Bag with 3 colored balls (red=1 pt, blue=2 pt, green=3 pt), equal chance to pick one.

- **i. Arithmetic mean of the points assigned to the balls:** the simple mean of the assigned values $(1, 2, 3)$:

$$\text{arithmetic mean} = \frac{1+2+3}{3} = \frac{6}{3} = 2.$$

- **ii. Expected value of the points when picking one ball at random (each $1/3$):**

$$E[\text{points}] = \tfrac{1}{3}\cdot 1 + \tfrac{1}{3}\cdot 2 + \tfrac{1}{3}\cdot 3 = 2.$$

So both are **2 points** (because probabilities are uniform).

Q6 — Fair six-sided die (faces 1..6). Arithmetic mean of face values and expected value on a roll.

- **i. Arithmetic mean of face values:**

$$\frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

- **ii. Expected value of points when you roll the die (each face probability $1/6$):**

$$E = 3.5.$$

So the expected score on one fair die roll is **3.5**.

Q7 — Class: 60% have completed assignment. You pick 3 students at random. Given that at least one has completed, probability all three have completed?

Let $p = 0.6$. For three independent picks:

- $P(\text{all three}) = p^3 = 0.6^3 = 0.216.$
- $P(\text{at least one}) = 1 - P(\text{none}) = 1 - (1-p)^3 = 1 - 0.4^3 = 1 - 0.064 = 0.936.$

Conditional probability:

$$P(\text{all three} \mid \text{at least one}) = \frac{0.216}{0.936} = \frac{216}{936} = \frac{3}{13} \approx 0.230769.$$

Answer: $3/13 \approx 0.23077$ (≈ 23.08%).

Q8 — 70% succeed on first attempt. Five players chosen; given at least one succeeded, probability all five succeeded?

Let $p = 0.7$. Then:

- $P(\text{all five}) = p^5 = 0.7^5 = 0.16807.$
- $P(\text{at least one}) = 1 - (1-p)^5 = 1 - 0.3^5 = 1 - 0.00243 = 0.99757.$

Conditional probability:

$$P(\text{all five} \mid \text{at least one}) = \frac{0.16807}{0.99757} \approx 0.1685.$$

Answer: approximately **0.1685** (≈ 16.85%).

Q9 — Party: 40% wear hats. You spot three people; the first one is wearing a hat. What's probability the other two are also wearing hats?

If people are sampled (effectively) independently (large population or sampling with replacement), then knowing the first has a hat doesn't change the marginal probabilities for the others. Thus

$$P(\text{2nd and 3rd both hat} \mid \text{1st hat}) = P(\text{2nd hat}) \cdot P(\text{3rd hat}) = 0.4 \times 0.4 = 0.16.$$

Answer: **0.16** (16%).
(If the population is small and sampling is without replacement you'd use hypergeometric probabilities; here independence/large population is the usual classroom assumption.)

Q10 — Two independent switches S1,S2 (each ON w.p. 1/2), connected in parallel. Probability lamp is ON?

Parallel means lamp is ON if **either** switch is ON (logical OR). Easier to compute via complement:

$$P(\text{lamp ON}) = 1 - P(\text{both OFF}) = 1 - \left(\tfrac{1}{2} \times \tfrac{1}{2}\right) = 1 - \tfrac{1}{4} = \tfrac{3}{4} = 0.75.$$

Answer: **3/4** (75%).

Quick summary table

- Q1: **XOR** has highest entropy (1 bit). OR/AND ≈ 0.8113 bits.
- Q2: Max entropy = uniform $p_i = 1/6$ (1 way), $H \approx 2.585$ bits. Min entropy = degenerate (6 ways), $H = 0$.
- Q3: $1/20 = 5\%$.
- Q4: **0** (exact instant in continuous uniform).
- Q5: arithmetic mean = expected value = **2**.
- Q6: arithmetic mean = expected value = **3.5**.
- Q7: $P(\text{all three} \mid \text{at least one}) = 3/13 \approx 0.2308.$
- Q8: $\approx 0.1685.$
- Q9: $0.16.$
- Q10: $0.75.$

If you want, I can:

- show short probability trees for Q5/Q6/Q7–Q10,
- produce small diagrams for Q1 (truth-tables + barplots of output probabilities), or
- give the exact hypergeometric correction for Q8 ↓ for finite-population sampling. Which would you like?